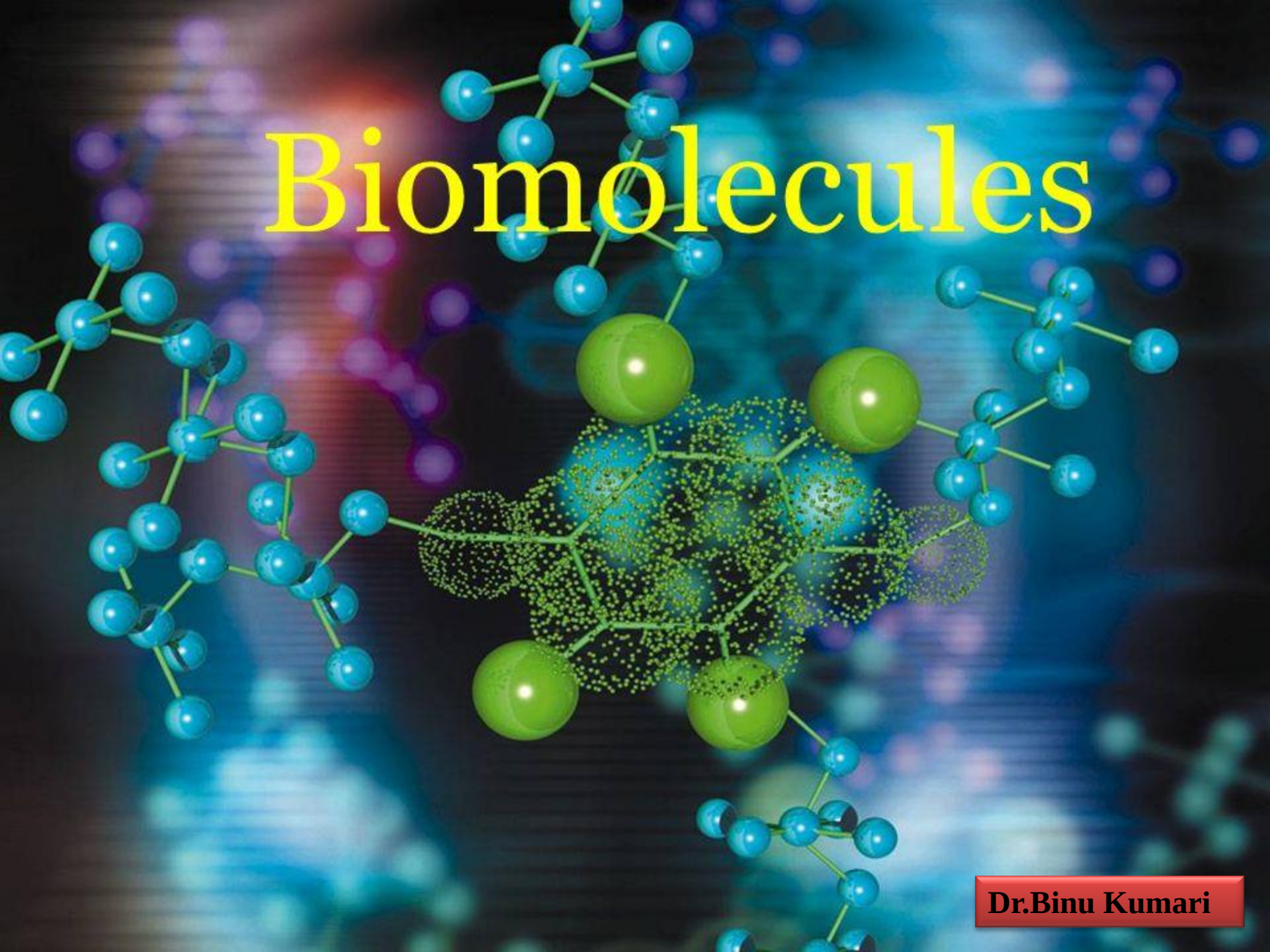


Biomolecules

The image features a complex 3D molecular model of a biomolecule. The central part of the molecule is a dense, green, textured cluster, possibly representing a metal core or a specific functional group. This central cluster is surrounded by several large, smooth, green spheres. Extending from these are numerous smaller, blue spheres connected by thin green lines, representing the rest of the molecule's structure. The background is a dark, blue, and purple gradient with a bokeh effect of light spots, giving it a scientific and futuristic feel.

Biomolecules

- Biomolecules are the most essential organic molecules, which are involved in the maintenance and metabolic processes of living organisms.
- Biomolecules are biological molecules produced by the cells of the living organism. They are critical for life as it helps organisms to carry out basic biological processes such as reproduction, growth and sustenance.
- There are four major classes of Biomolecules – Carbohydrates, Proteins, Nucleic acids and Lipids.

Carbohydrates

Carbohydrates are defined as **polyhydroxy** aldehydes or ketones compounds that produce biomolecule units on hydrolysis and are optically active. A few carbohydrates are also sweet in taste and are called sugars. These carbohydrates are classified based on their behaviour during hydrolysis.

Monosaccharides:

Carbohydrates that cannot be hydrolyzed further into simpler, singular units are called monosaccharides.

Examples : Glucose, Fructose, Ribose, etc.

Oligosaccharides:

Carbohydrates that produce two to ten monosaccharide units on hydrolysis are called oligosaccharides. For Example, if one molecule of sucrose is hydrolyzed, it will give one molecule of glucose and one molecule of fructose.

Polysaccharides:

Carbohydrates that produce more than ten monosaccharide units on hydrolysis are called polysaccharides. These are also known as non-sugars, as they do not taste sweet. Examples are Cellulose, starch, etc.

Carbohydrates:

The most abundant organic molecules in nature

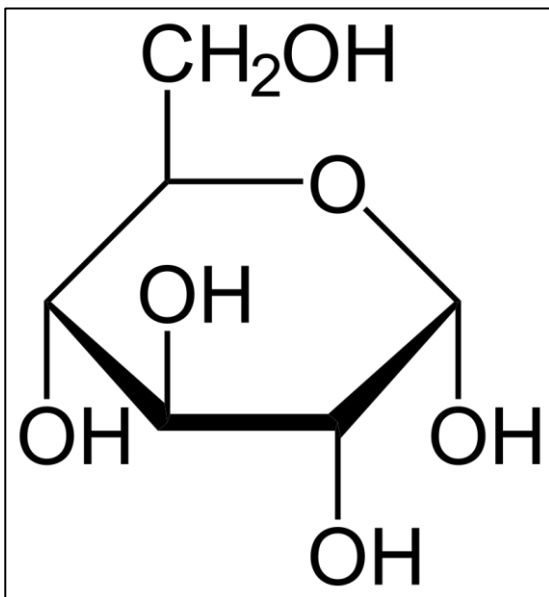
The empiric formula is $(\text{CH}_2\text{O})_n$, “hydrates of carbon”

Carbohydrates:

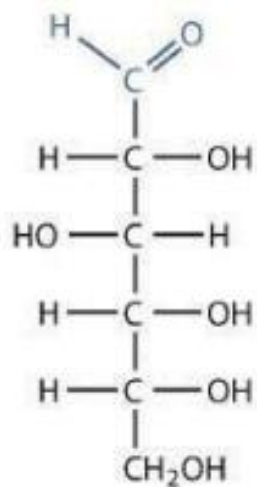
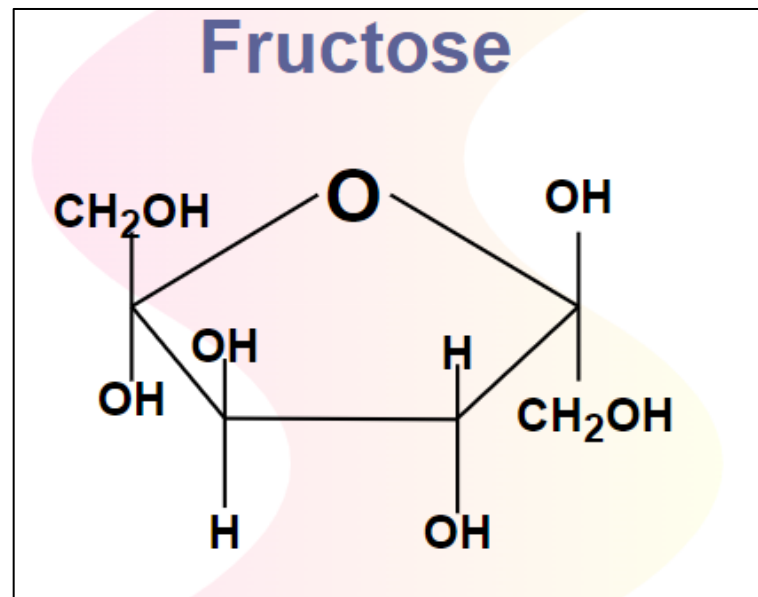
provide important part of energy in diet

Act as the storage form of energy in the body

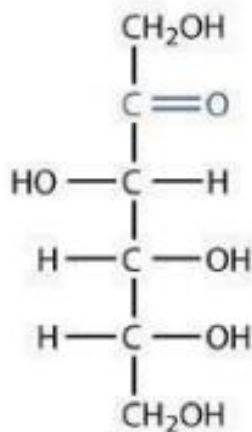
are structural component of cell membranes



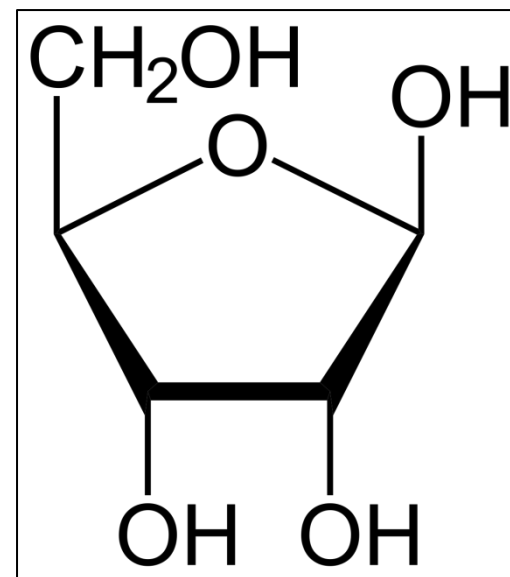
Glucose



Glucose
(an aldohexose)



Fructose
(a ketohexose)



Structure of Ribose

- **Many diseases associated with disorders of carbohydrate metabolism including:**

Diabetes mellitus

Galactosemia

Glycogen storage diseases

Lactose intolerance

Classification

- **Monosaccharides**: Simple sugar
- **Disaccharides**: 2 monosaccharide units
- **Oligosaccharides**: 3-10 monosaccharide units
- **Polysaccharides**: more than 10 sugar units
 - Homopolysaccharides &

Monosaccharides

Further classified based on:

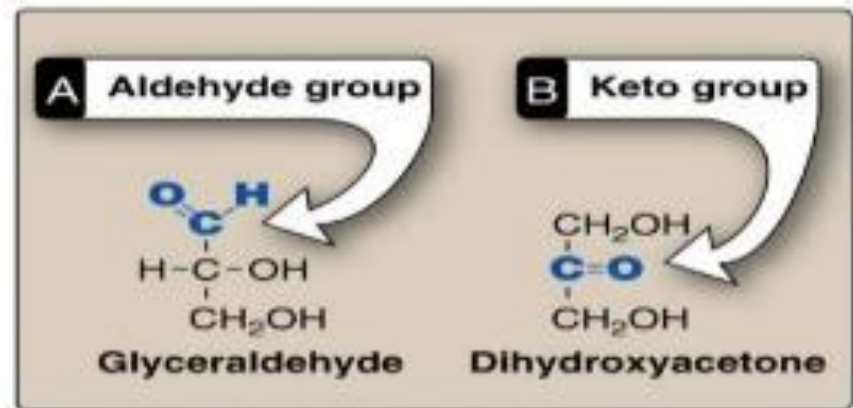
1. No. of carbon atoms

<u>Generic names</u>	<u>Examples</u>
3 carbons: trioses	Glyceraldehyde
4 carbons: tetroses	Erythrose
5 carbons: pentoses	Ribose
6 carbons: hexoses	Glucose
7 carbons: heptoses	Sedoheptulose
9 carbons: nonoses	Neuraminic acid

2. Functional sugar group:

Aldehyde group –
aldoses

Keto group –
ketoses



Monosaccharides

CONT'D

	Aldose	Ketose
Triose	Glyceraldehyde	Dihydroxyacetone
Pentose	Ribose	Ribulose
Hexose	Glucose	Fructose

Disaccharides

- **Joining of 2 monosaccharides
by O-glycosidic bond:**

Maltose (α -1, 4) = glucose + glucose

Sucrose (α -1,2) = glucose +
fructose

Lactose (β -1,4) = glucose +
galactose

Polysaccharides

□ Homopolysaccharides:

Branched:

Glycogen and starch (α -glycosidic polymer)

Unbranched:

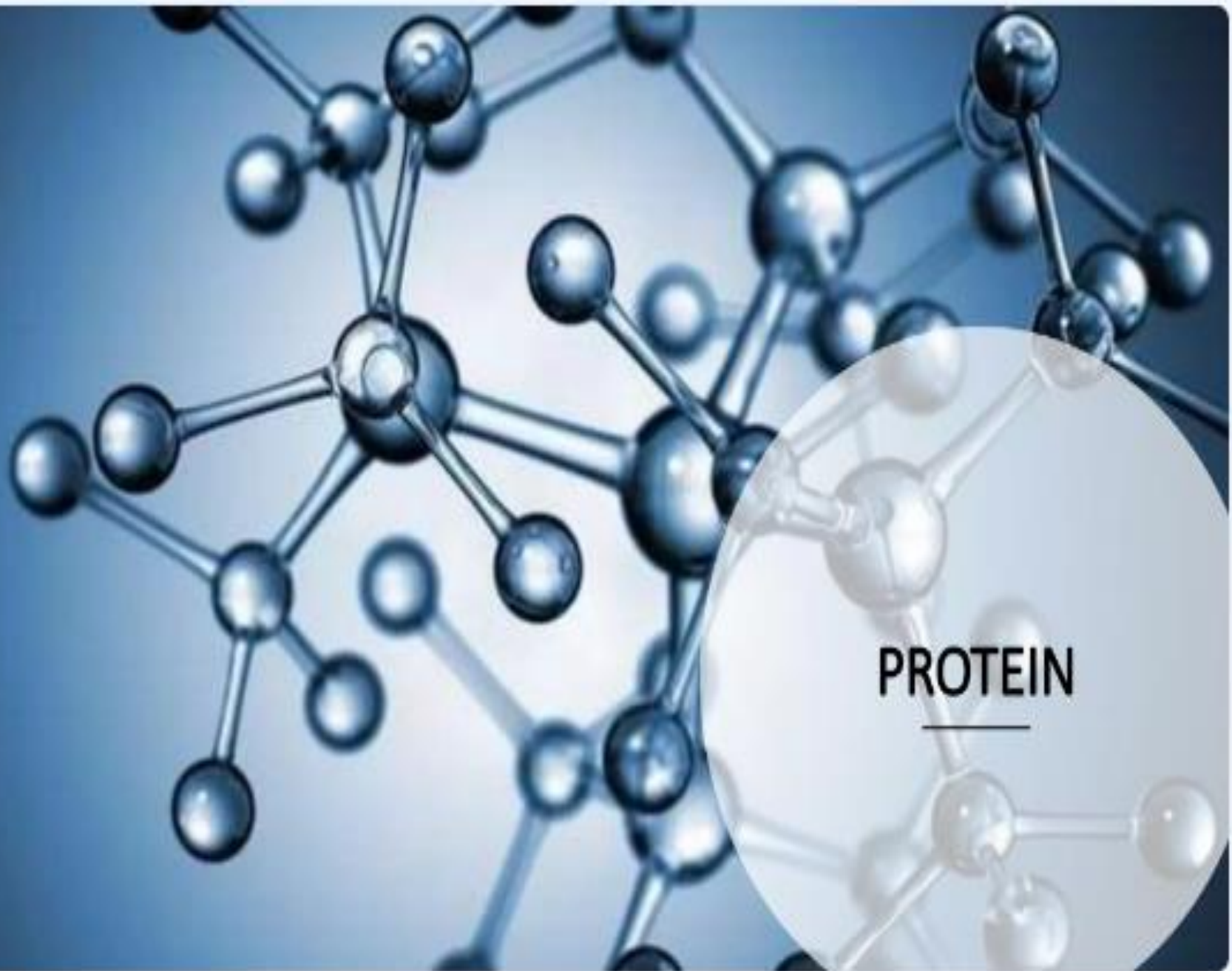
Cellulose (β -glycosidic polymer)

□ Heteropolysaccharides:

e.g., glycosaminoglycans (GAGs)

Carbohydrates

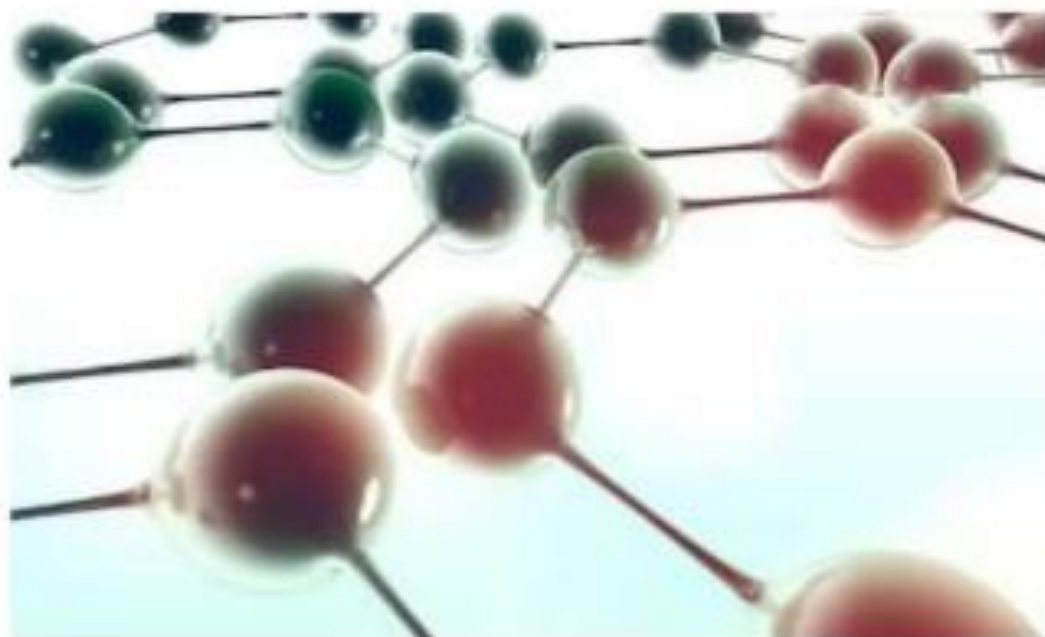
- They are collectively called as saccharides (Greek: sakcharon = sugar).
- Depending on the number of constituting sugar units obtained upon hydrolysis, they are classified as monosaccharides (1 unit), oligosaccharides (2-10 units) and polysaccharides (more than 10 units).
- They have multiple functions' viz. they're the most abundant dietary source of energy; they are structurally very important for many living organisms as they form a major structural component, e.g. cellulose is an important structural fibre for plants.



Proteins

- Proteins are another class of biomolecules, which make up around 50per cent of the cellular dry weight.
- Proteins are polymers of amino acids arranged in the form of polypeptide chains.
- The structure of proteins is classified as primary, secondary, tertiary and quaternary in some cases.
- These structures are based on the level of complexity of the folding of a polypeptide chain.
- Proteins play both structural and dynamic roles. Myosin is the protein that allows movement by contraction of muscles.
- Most enzymes are proteinaceous in nature.

Protein



- Protein, highly complex substance that is present in all living organisms. Proteins are of great nutritional value and are directly involved in the chemical processes essential for life.
- Jöns Jacob Berzelius, who in 1838 coined the term protein, a word derived from the Greek *proteios*, meaning "holding first place."

Structure

Protein structure is the three-dimensional arrangement of atoms in an amino acid-chain molecule. Proteins are polymers – specifically polypeptides – formed from sequences of amino acids, the monomers of the polymer. A single amino acid monomer may also be called a residue indicating a repeating unit of a polymer.

Proteins form by amino acids undergoing condensation reactions, in which the amino acids lose one water molecule per reaction in order to attach to one another with a peptide bond. By convention, a chain under 30 amino acids is often identified as a peptide, rather than a protein.

- Protein structures range in size from tens to several thousand amino acids. By physical size, proteins are classified as nanoparticles, between 1–100 nm. Very large aggregates can be formed from protein subunits. For example, many thousands of actin molecules assemble into a microfilament.
- A protein generally undergo reversible structural changes in performing its biological function. The alternative structures of the same protein are referred to as different conformational isomers, or simply, conformations, and transitions between them are called conformational changes.



Protein Function

transport material around the cell by locomotive and structural proteins (e.g. myosin and actin),

transport of material in and out of the cell (e.g. permeases and ion pumps),

locomotion of the cell (e.g. flagellar proteins),

regulate the access to and reading of genetic material for protein expression and replication of DNA (e.g. histones, repressors like LacI, and gyrase),

cellular energy homeostasis (e.g. ATP synthase and adenylate kinase),

cellular metabolism (interconversion of chemical intermediates in metabolic pathways by enzymes)

Nucleic Acids

- Nucleic acids refer to the genetic material found in the cell that carries all the hereditary information from parents to progeny.
- There are two types of nucleic acids namely, deoxyribonucleic acid (DNA) and ribonucleic acid (RNA).
- The main function of nucleic acid is the transfer of genetic information and synthesis of proteins by processes known as translation and transcription.
- The monomeric unit of nucleic acids is known as nucleotide and is composed of a nitrogenous base, pentose sugar, and phosphate.
- The nucleotides are linked by a 3' and 5' phosphodiester bond.
- The nitrogen base attached to the pentose sugar makes the nucleotide distinct.
- There are 4 major nitrogenous bases found in DNA: adenine, guanine, cytosine, and thymine.
- In RNA, thymine is replaced by uracil.
- The DNA structure is described as a double-helix or double-helical structure which is formed by hydrogen bonding between the bases of two antiparallel polynucleotide chains. Overall, the DNA structure looks similar to a twisted ladder.

Nucleic Acids

Nucleic acids, built by polymerizing nucleotides, function primarily as informational molecules for the storage and retrieval of information about the primary sequence of polypeptides.

Store and Retrieve Genetic Information

Nucleotide

- A nucleotide is the basic building block of nucleic acids (RNA and DNA).
- A nucleotide consists of a sugar molecule (either ribose in RNA or deoxyribose in DNA) attached to a phosphate group and a nitrogen-containing base.
- The bases used in DNA are adenine (A), cytosine (C), guanine (G) and thymine (T).



Nucleic Acids

DNA

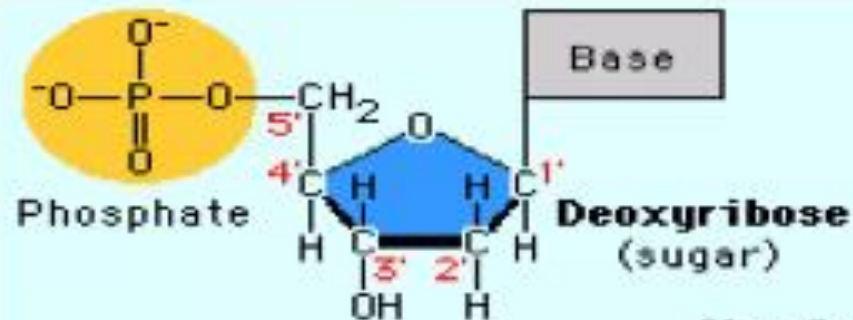
Deoxyribonucleic acid serves as a cellular database by storing an immense amount of information about all the polypeptides a cell can potentially make

RNA

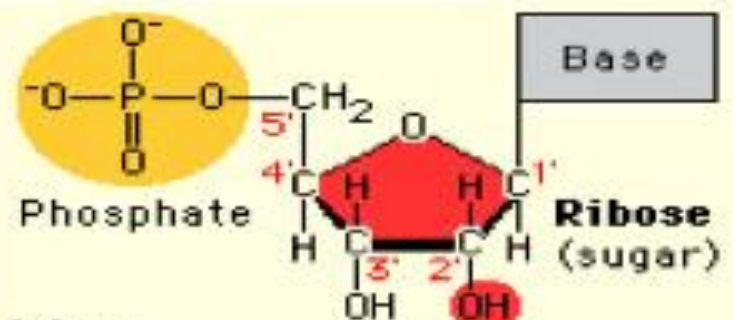
Ribonucleic acid occurs in several different forms (messenger RNA, ribosomal RNA, transfer RNA) and is needed to convert DNA information into polypeptide sequences. In some viruses, RNA serves as the primary database with no DNA involvement.

DNA and RNA

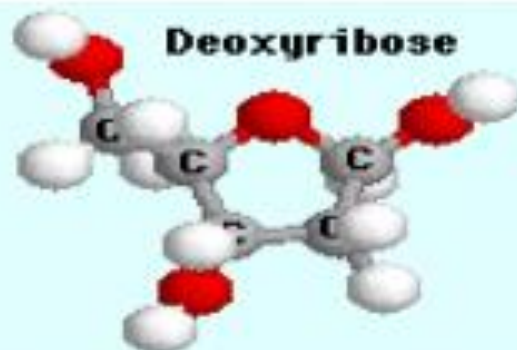
DNA



RNA



Nucleotides



Sugars

Nucleotides



- **Basic unit of DNA**
- **Three Parts**

Deoxyribose sugar

Phosphoric acid

Nitrogen base

Phosphate



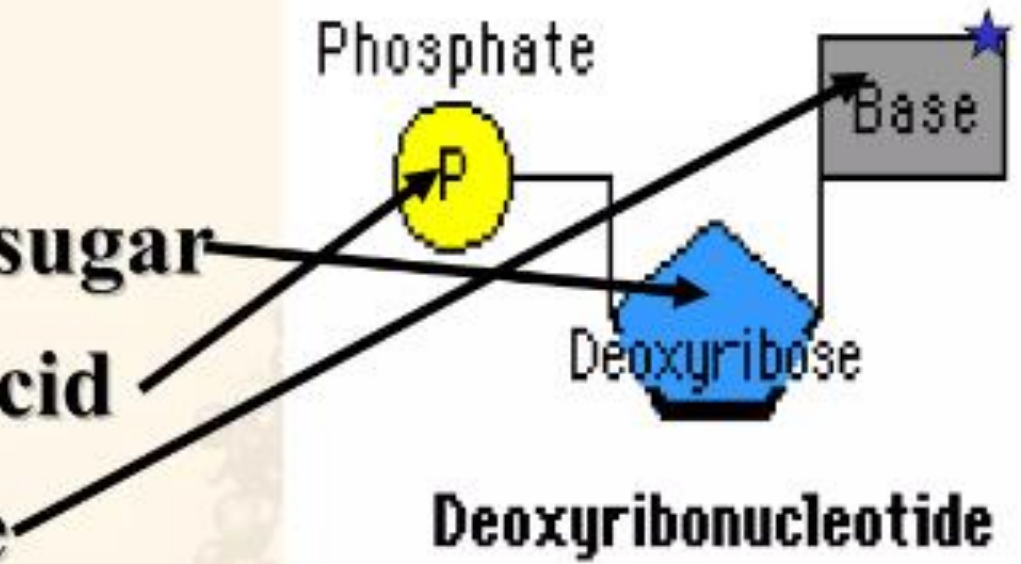
Deoxyribose



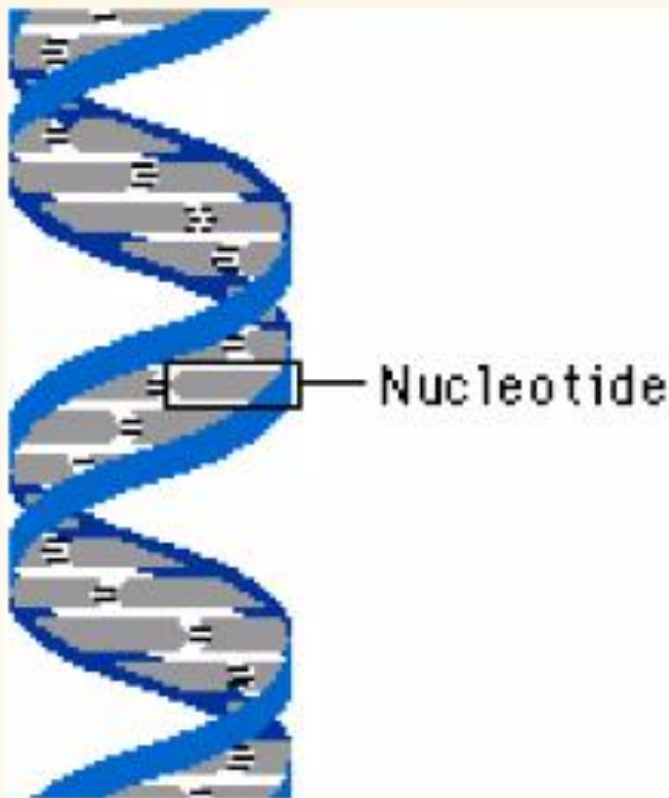
Base



Deoxyribonucleotide



Nucleotides



**Many nucleotides
are pieced
together to make a
DNA molecule**

Lipids

- Lipids are organic substances that are insoluble in water, soluble in organic solvents, are related to fatty acids and are utilized by the living cell.
- They include fats, waxes, sterols, fat-soluble vitamins, mono-, di- or triglycerides, phospholipids, etc.
- Unlike carbohydrates, proteins, and nucleic acids, lipids are not polymeric molecules. Lipids play a great role in the cellular structure and are the chief source of energy.

Important question

1. What are carbohydrates? Draw the structure of glucose and fructose.
2. What is biomolecules? Write 4 characteristics of biomolecules.
3. What are the 4 main classes of biomolecules?
4. What is glycogen?
5. What is nucleotides?